Loom



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **Medium** 3 labs

Covers these lab items: Bamboo Mat Weaving Loom; Handloom Set Up; Handloom/Power loom

What it is used for

A loom holds warp threads under controlled tension while weft threads are interlaced to produce woven fabric, mats, or other textile products. Correct setup and operation improve weave quality, productivity, and operator safety.

Main parts

- Loom frame and supporting structure
- Warp beam and cloth beam
- Heddles, harnesses, or thread guides
- Reed and beater mechanism
- Shuttle or weft insertion system
- Tension controls, pedals, or drive mechanism depending on loom type

Before you start

1. Identify whether the loom is hand-operated, power-driven, or designed for bamboo mat weaving.
2. Read the operating instructions and confirm the permitted material, width, and production settings.
3. Inspect the frame, beams, heddles, reed, shuttle, drive system, and guards for damage.
4. Check that the loom is stable, correctly installed, and clear of obstructions.
5. Arrange warp threads in the required order and secure them to the warp beam.
6. Set thread tension evenly and check that the weft material feeds without snagging.

How to use

1. Set the required fabric or mat width using the loom's approved adjustment method.
2. Thread the warp through the designated guides, heddles, and reed according to the weave pattern.
3. Secure the warp ends to the cloth beam and verify even tension across the width.
4. Load the shuttle or prepare the approved weft insertion mechanism.
5. For handlooms, operate the pedals or levers in the correct sequence to form the shed.
6. Insert the weft across the warp and use the beater to pack it evenly.
7. For power looms, start the drive using the designated control and observe the initial operation.
8. Inspect the developing weave for gaps, uneven tension, broken threads, or pattern errors.
9. Stop the loom before correcting thread breaks, jams, or alignment problems.

10. At completion, stop and isolate the drive if fitted, remove the woven material, and clean the work area.

⚠ Safety

- Keep fingers, hair, and loose clothing away from moving shafts, gears, belts, and the beater.
- Never reach into a moving loom to repair broken threads or remove a jam.
- Keep guards fitted on powered drives and do not bypass safety switches.
- Use safe lifting methods when handling heavy beams or material rolls.
- Switch off and isolate power before cleaning, adjustment, or maintenance on powered looms.
- Keep the floor clear of loose threads, bamboo strips, and other tripping hazards.

Care & storage

- Remove lint, loose threads, and material debris after use.
- Inspect warp beams, heddles, reed, bearings, belts, and drive components regularly.
- Lubricate approved moving parts according to the manufacturer's instructions.
- Store weaving materials in a clean, dry place and report damaged components promptly.

Common mistakes to avoid

- Uneven warp tension can produce distorted fabric or mats.
- Incorrect threading through the heddles or reed can create weave defects.
- Operating with damaged guards or loose drive components increases injury risk.
- Trying to clear a jam while the loom is moving can cause serious hand injuries.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.